Replicator dynamics

The room is the population

Split
Dig or Stay. Stand on your side.

Select
The fitter side gains a few.

Settle
Watch where it stops moving.

In the room: [ ]

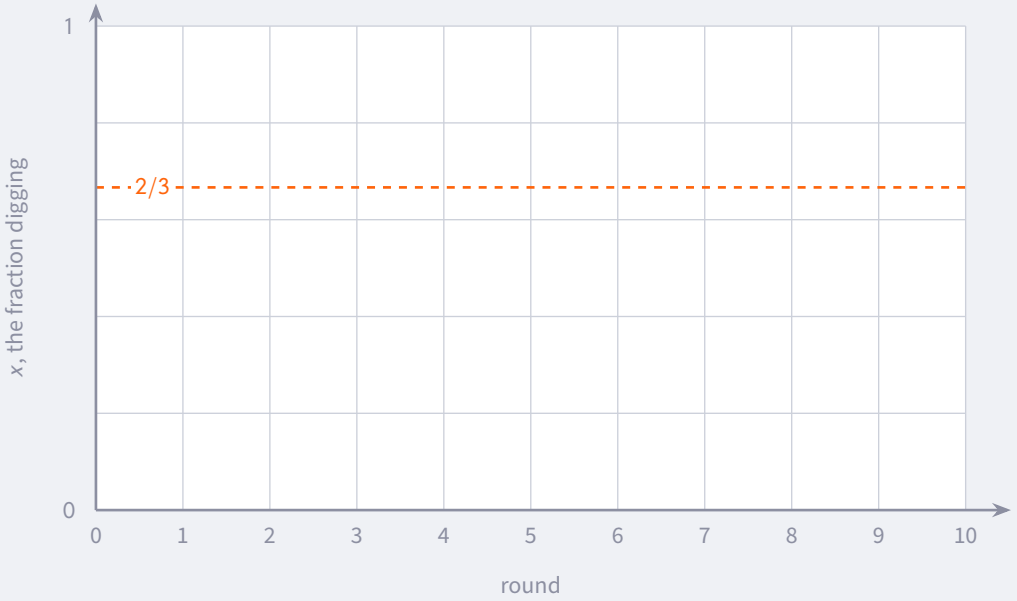
Starting split: [ ] digging

	Dig	Stay
Dig	3	2
Stay	4	0

Clearing the drift is worth 4 to each of you. Digging costs 2, shared if you both dig.

Two diggers get 3 each. A lone digger gets _____ while the other free-rides for _____. Two stayers get _____.

Would you rather be the one digging? _____



Round	x	f_D	f_S
1			
2			
3			
4			
5			
6			

Plot the split each round. The dashed line is where we expect it to end up.

With a fraction x digging,

$f_D =$ $f_S =$

They are equal when $x =$, which is where the room _____.

A strategy grows when it does better than the population average $\phi = xf_D + (1-x)f_S$. How fast it grows depends on two things:

how far above average it is: ____ and how many already play it: ____

$$\dot{x} = ____$$

Each round of the activity, with its small fixed step, was one step of _____ on this equation.

Factorise $\dot{x}$ and read off every value where it is zero:

Rest points: [ ] [ ] [ ]

For $0 < x < 2/3$ the sign of $\dot{x}$ is [ ], and for $2/3 < x < 1$ it is [ ]. So from anywhere inside, the population moves towards [ ].

A small group of Stayers invades a population sitting at $x = 2/3$. Do they earn more or less than the residents?

So $x = 2/3$ _____ an ESS. Rock-Paper-Scissors has an interior rest point too, but there the dynamics _____ instead of settling.

What we just did in the room is **Question 1** on the Replicator Dynamics page.

(a)	The fitnesses f_D and f_S as functions of x.	[3]
(b)	The replicator equation, simplified to a product of factors.	[7]
(c)	All stable populations.	[3]
(d)	From the sign of $\dot{x}$, which one we approach, and where the class settled.	[6]
(e)	Whether the interior population is an ESS, argued two ways.	[6]

vknight.org/gt/topics/replicator-dynamics.html